

ABASHIRI, Japan

WMO: 474090

Lat: 44.0178N

Lon: 144.2797E

Elev: 143

StdP: 14.62

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.0	10.9	-0.7	5.3	12.1	2.0	6.1	13.9	28.0	25.5	25.3	24.9	6.3	240	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.5	82.3	71.3	78.8	69.4	75.8	67.7	73.3	80.0	71.3	76.8	69.4	74.3	8.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.8	114.1	78.0	69.1	107.6	75.6	67.5	101.5	73.4	37.1	80.3	35.3	77.0	33.7	74.4	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	18.9	16.3	DB	3.0	90.9	3.3	4.1	0.7	93.8	-1.2	96.2	-3.1	98.6	-5.4	101.5	(4)
(5)			WB	2.2	76.2	3.1	1.8	-0.1	77.5	-1.9	78.6	-3.6	79.6	-5.9	80.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	21.9	21.6	29.5	40.5	50.2	56.7	64.2	67.9	62.7	51.8	39.2	27.4	(6)	
	DBStd	17.35	5.47	6.44	6.54	7.51	8.63	7.25	6.73	5.95	5.39	5.77	7.00	5.78	(7)	
	HDD50	3807	870	795	635	303	107	18	0	0	0	48	330	701	(8)	
	HDD65	7705	1335	1215	1100	736	465	262	98	36	107	410	775	1166	(9)	
	CDD50	1837	0	0	0	17	112	220	442	555	382	104	5	0	(10)	
	CDD65	259	0	0	0	0	5	14	74	126	39	1	0	0	(11)	
	CDH74	1088	0	0	0	10	39	59	316	533	129	2	0	0	(12)	
	CDH80	230	0	0	0	2	9	7	72	119	21	0	0	0	(13)	
(14) Wind	WSAvg	7.3	8.6	7.9	8.0	7.9	7.1	5.7	5.4	5.7	6.6	7.7	8.1	8.5	(14)	
(15) Precipitation	PrecAvg	33.2	2.4	1.5	2.0	2.0	2.5	2.9	3.3	4.3	4.2	3.2	2.5	2.3	(15)	
	PrecMax	47.6	5.8	4.4	5.6	6.3	6.7	8.2	8.3	16.7	11.8	8.3	5.2	6.9	(16)	
	PrecMin	22.8	0.4	0.3	0.5	0.3	0.7	0.4	0.4	0.6	0.4	0.9	0.4	0.6	(17)	
	PrecStd	5.6	1.0	0.9	1.2	1.1	1.3	1.6	1.7	2.7	2.2	1.7	1.1	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.0	43.6	52.0	71.8	78.8	80.0	87.7	88.1	83.0	70.8	61.9	48.3	(19)	
		MCWB	35.8	37.4	42.5	53.2	59.0	65.0	74.7	75.2	71.7	59.2	52.4	43.4	(20)	
	2%	DB	34.9	37.8	45.9	61.9	72.3	74.7	80.7	83.1	77.6	66.5	57.1	42.4	(21)	
		MCWB	31.3	32.7	39.1	48.8	56.3	62.8	71.0	73.2	68.3	57.3	49.6	37.9	(22)	
	5%	DB	32.3	34.2	42.5	56.8	67.4	71.0	77.1	79.5	73.8	63.2	53.4	38.9	(23)	
		MCWB	29.7	30.2	36.1	46.2	55.3	61.4	69.0	71.3	66.2	55.9	47.0	35.1	(24)	
	10%	DB	30.4	31.2	39.1	52.1	63.2	67.7	74.0	76.4	70.8	60.7	49.7	36.1	(25)	
		MCWB	28.0	28.1	33.9	43.7	53.4	59.7	67.5	69.6	64.2	54.4	44.4	32.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.1	38.1	43.2	54.9	61.4	67.2	75.3	76.6	73.3	62.4	54.7	44.3	(27)	
		MCDB	39.5	42.6	49.5	67.7	74.2	76.2	86.3	86.5	80.6	67.2	60.0	47.5	(28)	
	2%	WB	32.2	33.0	39.7	50.3	58.5	64.5	72.2	74.1	70.4	59.3	50.5	38.6	(29)	
		MCDB	34.0	36.9	45.5	60.1	68.9	72.8	79.0	81.3	75.4	64.2	55.2	41.6	(30)	
	5%	WB	30.1	30.6	36.7	46.8	56.2	62.3	70.1	72.2	67.7	57.2	47.5	35.7	(31)	
		MCDB	31.8	33.7	41.7	55.2	65.8	69.2	75.7	77.8	71.5	62.1	52.5	38.3	(32)	
	10%	WB	28.2	28.4	34.2	44.3	54.0	60.4	68.2	70.6	65.2	55.1	44.8	32.9	(33)	
		MCDB	30.3	31.0	38.7	51.3	62.2	66.7	73.2	75.5	69.8	59.8	49.4	35.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	12.0	11.6	13.8	13.8	12.0	10.9	10.5	11.6	12.9	11.6	10.5	(35)	
		MCDBR	12.7	16.5	17.1	23.6	22.5	19.1	16.7	15.6	15.4	16.0	17.6	13.7	(36)	
	5% WB	MCWBR	10.9	12.9	12.0	13.1	11.4	10.0	8.7	8.1	8.9	10.3	12.7	12.0	(37)	
		MCWBR	11.4	15.3	16.6	22.1	20.1	16.4	15.1	14.2	13.3	14.1	16.1	13.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.332	0.373	0.445	0.507	0.522	0.499	0.481	0.460	0.408	0.398	0.373	0.344	(40)	
		taud	2.074	1.898	1.818	1.828	1.887	2.048	2.166	2.242	2.338	2.218	2.192	2.118	(41)	
	Ebn at Noon		238	246	244	241	242	249	252	253	258	240	223	220	(42)	
		Edh at Noon	33	47	59	63	61	52	46	41	35	34	30	29	(43)	
(44) All-Sky Solar Radiation	RadAvg		516	841	1196	1449	1549	1577	1476	1342	1151	859	537	423	(44)	
	RadStd		44	72	88	100	134	137	162	97	90	41	44	32	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page